Aidan Kennedy

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Internships & Experience

Darkvision Technologies. • Mechanical Design Co-op• *January 2026– July 2026*

- Designed a 3-axis micro-CNC for high-precision microfabrication, capable of drilling 25 µm holes with 3 µm, complete with an electrical control cabinet, custom mechanical design, integrated coolant circulation system, and adapted microstage firmware. This project reduced part manufacturing time from ~40 hours on a full-size CNC to 6 hours. This project became cost-effective after 8 machined parts and has already achieved this.
- Set up odometer electrical and firmware test infrastructure to root-cause ILI field data issues, measuring wheel slip/stall in multiple conditions

H3X Technologies. • Mechanical Design Co-op• *September 2025– January 2025*

- Responsible for mechanical design of HV-DC and Inverter area assembly on HPDM-500 product
- Extensive work in Ansys mechanical and vibration analysis as well as Solidworks GD&T drawings
- Designed a rotating test fixture for shaft assemblies to test pressurized seals at altitude and bearing lifetime analysis with variable radial loads to research lifetime analysis and thermal generation
- Designed HPDG-30 Genset mount with ANSYS Modal analysis for vibration validation in representative conditions

TESLA MOTORS. • Mechanical Design Co-op • *Jan 2025 – September 2025*

- Designed DC High Potential test to examine semiconductors for mechanical failures, using MATLAB to complete a full sweep test capable of reacting to different failure conditions and automatically saving data to our cloud
- Root cause analysis of semiconductor failure due to pressure applied through mechanical constraints using MATLAB controlled paired compression and electrical testing to inform inverter redesign specifications
- Designed custom PCBA for use in HTHE (85degC 85% humidity) chamber for lifetime modelling of inverter components at high voltages, measuring time to failure with an external Data Acquisition System

BORON ENERGY CO. • Mechanical Design Co-op • *May 2024 – Sept 2024*

- Designed a convection-cooled heat sink using Altair suite (CFD and FEA) simulations to optimize the heat sink design, incorporating tolerance stack-up and GD&T drawings to manage a 0.3mm gap to PCBA components for optimal heat removal, while also designing for high-volume production of cast materials
- Manufactured and designed electric drivetrain dynamometer capable of measuring inverter, motor, or full system 3D efficiency maps by measuring currents, voltages, and mechanical output at all stages of the drivetrain
- Developed python simulation to define exact system accuracy of the dyno DAQ throughout full three-dimensional power spectrum to be paired with efficiency maps to provide exact localized accuracy

Design Team

UBC FORMULA ELECTRIC DESIGN TEAM • *September 2022 – June 2025*

Mechanical Technical Director • *April 2024 – June 2025*

- Co-leading a University team of over 80 students designing and building a Formula-style electric racecar
- Set up a top-down parametric Solidworks CAD design infrastructure to enable more efficient CAD design
- Quad motor integrated cooling loop, custom power electronics cooling plates, and motor cooling sleeves to optimize component temperature during our competitions 22Km endurance event
- Manufacturing two-stage compound planetary gearboxes for in-hub development producing 300Nm per wheel

Drivetrain Sub-Team Lead • *August 2023 – March 2024*

- Designed positive pressure battery pack cooling system, graded to IP-54 specifications, used to uniformly cool 6 separate battery pack segments, using SolidWorks, FEA flow simulations, fabricated metal and 3D-printed parts
- Characterized, designed, and tested a liquid cooling loop for our electric drivetrain with wind tunnel testing, lap simulation, and efficiency testing to keep our motors and inverters at optimal temperatures during endurance events with high ambient air temperatures of 30-40 degrees, and an 80KW peak output powertrain

Skills

MECHANICAL DESIGN

- SOLIDWORKS
- FEA/CFD Simulation
- Parametric Design

MANUFACTURING

- Milling
- Lathe
- Carbon Fiber Layup

OTHER

- MATLAB
- Python
- Atlassian Suite

Edu

University of British Columbia • *Sept 2023 – May 2026*
Bachelor of Applied Science - Mechanical Engineering,

Kwantlen Polytechnic University • *Sept 2022 – May 2023*
Engineering Transfer Program - Deans Honor List